

BROOKE WINKLER

B.S. Mechanical Engineering & Aerospace Engineering · Honors Program · Michigan Technological University · Expected April 2030

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RACE OPERATIONS

01

Captain — Design, Strategy, Build & Electrical Lead

Aug 2022 – May 2026

FRC TEAM 7598 — SCA CONSTELLATIONS · WIXOM, MI

- Led mechanical design, fabrication, electrical systems, and competition strategy for a student-run FIRST Robotics Competition team.
- Served as the primary contributor for mechanical and electrical development during senior year, taking designs from CAD through fabrication, assembly, wiring, and competition use.
- Utilized SolidWorks for detailed mechanical design and 3D printing, CNC machining, and laser cutting for rapid prototyping and final components.
- Supported robot programming and controls development using Java and WPILib while independently tuning robot parameters and operational values.
- Led and mentored newer team members in mechanical design, fabrication, electrical work, strategy, and competition operations.
- Guided the team through regional and state-level competition and qualification for the FIRST World Championship.

Media Team — Photographer & Journalist

Jul 2026

RED BULL RACING DETROIT SHOWRUN · DETROIT, MI

- Supported media operations for the Red Bull Racing Detroit Showrun, gaining firsthand exposure to professional motorsports media and event operations.
- Served as a photographer and journalist, sharing experiences with local news outlets to promote events.

Business Development

Dec 2025 – Present

AVEXEL WEB DESIGN · HARBOR SPRINGS, MI

- Conduct client outreach through cold calling and email to develop business opportunities for a student-run web development company.
- Conduct market research to identify potential clients and evaluate local business needs and opportunities.
- Support marketing and communications efforts while collaborating with developers on client-facing initiatives.

STRATEGY & SIMULATION

02

2026 Formula 1 Power Unit Regulations — Independent Research

2026 – Present

POWERTRAIN, DEPLOYMENT & COMPETITIVE DYNAMICS · SELF-DIRECTED

- Conducting research into the 2026 Formula 1 power unit regulations and their effects on racing.
- Investigating changes in powertrain design, hybridization, vehicle performance, and competitive racing dynamics.

LEADERSHIP UNDER PRESSURE

03

Varsity Golf Captain

2022 – 2025

ST. CATHERINE OF SIENA ACADEMY · WIXOM, MI

- Served as varsity team captain during senior year after three years as a varsity athlete.
- Earned All-Catholic, All-League, and All-Academic honors.

Technical Director & Stage Manager

2023 – 2026

SCA DRAMA · WIXOM, MI

- Led backstage technical operations and crews across multiple productions, coordinating scene changes, props, logistics, and crew responsibilities during rehearsals and performances.
- Developed practical solutions to complex production and time-management challenges while maintaining show readiness under high-pressure conditions.
- **2x Superior Award — Stage Management**

Mentor & Volunteer

2022 – 2026

FIRST ROBOTICS — SOUTHEAST MICHIGAN · MICHIGAN

- Mentored two FIRST Tech Challenge teams, including helping establish a rookie team and supporting another team through qualification for the Michigan State Championship.
- Volunteered at FIRST robotics competitions and supported students and teams throughout competition operations.

EDUCATION

Michigan Technological University

Houghton, MI

St. Catherine of Siena Academy

High School Diploma

Wixom, MI

GPA 3.8 · May 2026

TOOLKIT

DATA & SIMULATION

MATLAB · Mathematica · Parameter tuning · Data analysis · Scouting & pattern reading

RACE OPERATIONS

Match strategy · Pit operations · Event operations · Technical briefing · Team leadership

CAD & DESIGN

SolidWorks · Onshape · Fusion 360 · AutoCAD

FABRICATION

3D printing · CNC machining · Laser cutting · Soldering · Mechanical fabrication

CONTROLS & CODE

Java · WPILib · JavaScript · Git / GitHub

HONORS

- 3x Top Technician in the State of Michigan
- NASA-funded scholarship recipient
- Renaissance Woman Award
- FIRST Impact Award · World Championship Qualifier

COURSEWORK

FIRST-YEAR, MICHIGAN TECH

Calculus I & II · University Physics · General Chemistry · Engineering Fundamentals · Technical Graphics / CAD

NEXT

Formula 1, IndyCar, IMSA or an OEM team — anywhere strategy and telemetry are taken seriously. Building toward Formula SAE at Michigan Tech and eager to put hands on a real data stack.

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